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Project Description

ArtConnect Application

Step 3: Database Implementation and Advanced Features

Objective: Create the database in the DBMS, populate it with sample data, and set up advanced database objects (views, indexes, triggers, stored procedures/functions).

1. Create and populate the database

Prompt used for the LLM to insert sample data.

>>> Please provide the insertion queries used to populate the database, whose creation script is as follows:

[DATABASE CREATION SCRIPT]

Foreign keys must refer to existing primary keys: provide the lines starting with filling in the tables in which there are no foreign keys, then the tables in which the foreign keys refer to primary keys in tables that have already been filled in.

Please make the data various and pertinent so that we can create and verify procedures, triggers, vues etc.

The number of insertions should be large.

Provide the set in the form of an SQL script ready to be executed.

>>> can you make this 3 or 4 times larger ?

>>> using the pdf i gave you, retrieve the enum types from the class diagram and update the insert so that it follows them

The resulting data can be found at `artProjectInsertData.sql`

2. Views, indexes, and access rights

Views

- View all information's on all workshops and their instructors
- All tables without ID
- View all the information about sold artworks

Found in views.sql

Indexes

- Index on the status of artwork
- Index on workshops price
- Index on workshops level

Found in indexes.sql

3. Triggers and stored programs

Triggers

On delete

- Cannot delete an account if it is registered to a workshop

On update

- If an artwork is sold, it no longer belongs to the gallery

On insert

- Verify the validity of a workshop date when it is created
- Verifying the price of an artwork is strictly positive
- Verify that there is place left before booking a workshop

Found in trigger.sql

Stored functions:

- Return the number of participants currently booked for a given workshop. This could be written as a procedure, but the goal is to have quick reports using SELECT instead of CALL
- Return the average review rating for a given artwork, or NULL if no reviews exist yet.
- Return 1 (TRUE) if the workshop has reached max capacity, 0 (FALSE) otherwise.

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Stored procedures

- When you sell an artwork, update it and mark as “sold”. This will trigger the trigger when an artwork is sold to say it doesn’t belong to a gallery anymore
- display workshops that will be held at a specific location
- display arts exhibitions that will be held at a specific location
- display arts exhibitions that have a special theme.

Found in stored_procedures_functions.sql

4. Transactions

Register to all beginner workshop in a city, if two have the same date-hour, then cancel the transaction.

Found in transaction.sql.